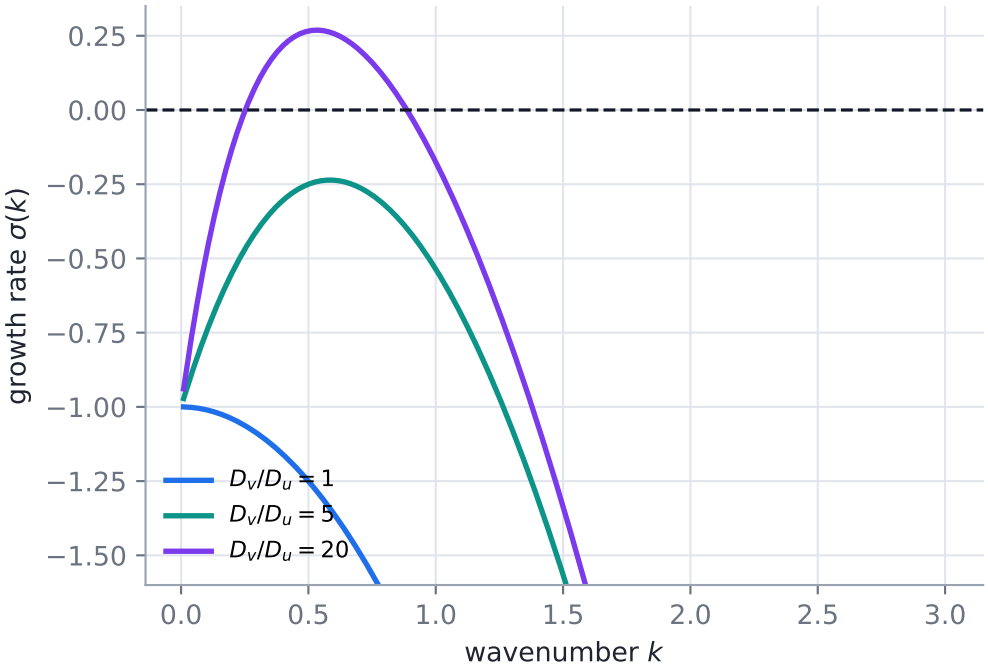
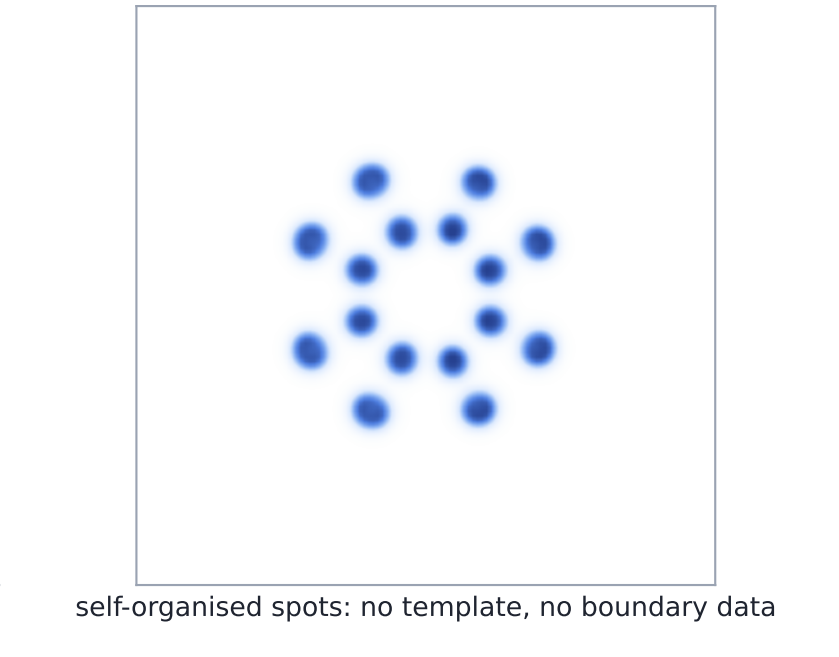

dispersion relation of a Turing instability
diffusion *destabilises*: only a band of k grows
growth rate $\sigma(k)$
wavenumber k
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
0.0
0.5
1.0
1.5
2.0
2.5
3.0
$D_v/D_u = 1$
$D_v/D_u = 5$
$D_v/D_u = 20$


Gray–Scott simulation on 128^2
the same nonlinearity that writes zebra stripes
self-organised spots: no template, no boundary data